

Working Memory Development: The Effects of Speech Rate, Word Length, and Acoustic Similarity on Serial Recall

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The effects of word length and acoustic similarity on speech rate and serial recall performance were investigated in children ranging in age from 4 to 10 years. Children of different ages show a word-length effect of equivalent size, and increases in recall with age are well accounted for by corresponding increases in speech rate. In contrast to this, the absolute difference in recall between acoustically similar and dissimilar lists increases as children get older, and the size of this difference is related to speech rate. It is argued that the pattern of results obtained may be explained in terms of a development of the articulatory loop as embodied in A. D. Baddeley and G. Hitch's (1974, in G. H. Bower, Ed., *The psychology of learning and motivation*, Vol. 8, New York: Academic Press) working memory model. © 1989 Academic Press, Inc.

In recent years research on short-term memory and its development has been prolific (for a recent review see Dempster, 1985). One impetus to such research has been the view that short-term memory may be an important component of a working memory system that holds verbal information in the course of more complex tasks such as reading, mental arithmetic, and reasoning (Baddeley, 1986; Baddeley & Hitch, 1974). In this view an understanding of short-term memory and its development could have considerable implications for understanding other aspects of cognitive development.

The present research is concerned with exploring the mechanisms responsible for improvements in short-term memory in early to middle childhood. There is a great deal of evidence linking verbal short-term memory to the use of a speech-based code (see, for example, Crowder, 1978; Dempster, 1985). Two effects are central to this evidence. First is the effect of word length. Baddeley, Thomson, and Buchanan (1975)

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showed that subjects are able to recall more short than long words in a serial recall task. The relationship between recall and the time taken by subjects to repeat the words used was in fact very simple; people recalled as much as they could say in a fixed time interval of about 1.8 s. Second is the effect of acoustic similarity; it is more difficult to remember lists of acoustically similar consonants or words in a serial recall task even when they are presented visually (Baddeley, 1966; Conrad, 1963; Conrad & Hull, 1964). A proper understanding of these two effects and of their interrelationships and developmental course is central to an understanding of short-term memory and its development.

Developmentally the effects of word length on short-term memory are quite straightforward. In some recent experiments we found that a common linear function relating recall of words of different lengths to speech rate fitted the results for subjects of widely differing ages (4 year olds to adults) (Hulme, Thomson, Muir, & Lawrence, 1984). Similar results for smaller ranges of ages have also been obtained by others (Hitch & Halliday, 1983; Nicholson, 1979).

The effects of acoustic similarity on recall at different ages and the interpretation of these effects is much less clear. It has recently been shown that memory span is progressively more influenced by acoustic similarity with increasing age (Hulme, 1984). Traditionally, the effects of word length and acoustic similarity have both been attributed to the operation of the same verbal short-term memory system (e.g., Baddeley & Hitch, 1974). More recently, however, the acoustic similarity effect has been attributed to the operation of a separate component of the articulatory loop, the phonological input store (Baddeley, 1986). It is not clear how either of these views can account for the fact that acoustic similarity has more effect on recall as children get older. The present experiments are designed to clarify this issue and to contribute to a more adequate understanding of the acoustic similarity effect and its development. A better understanding of the development of this effect should in turn be relevant to evaluating current models of short-term memory which are usually only evaluated on the basis of data from adult studies (Hulme, 1986).

The experiments which have examined the effects of word length on recall in children have shown that increases in memory span with age are paralleled by corresponding changes in the maximum rate at which the children can repeat the words which they are memorizing. This has been explained by hypothesizing that speech rate provides us with a measure of rehearsal rate in short-term memory. It is conceivable that the effects of acoustic similarity on recall depend upon differences in speech rate with sequences of acoustically similar items being harder to articulate rapidly. Another possibility is that the effects of acoustic similarity on recall depend upon the occurrence of confusions during rehearsal. If

the younger children speak, and so rehearse, material more slowly, this might account for why acoustic similarity has more effect on recall in older children. Testing these possibilities will provide further evidence relevant to the importance of changes in speech rate as an explanation for developmental differences in short-term memory performance. To provide such a test, recall and speech rate for acoustically similar and dissimilar words and for words of differing lengths were assessed in three groups of children ranging in age from 4 to 10 years.

EXPERIMENT 1

Method

Word Length

Subjects. A total of 36 children participated in the study; twelve 4 year olds (mean age 4 years 8 months; range 4 years 5 months to 4 years 11 months), twelve 7 year olds (mean age 7 years 9 months; range 7 years 7 months to 7 years 11 months), and twelve 10 year olds (mean age 10 years 10 months; range 10 years 8 months to 10 years 11 months). There were equal numbers of boys and girls in each age group. The children came from two different schools both of which had predominantly middle-class intakes.

Materials. In order to be able to relate recall to speech rate, measures of both variables were obtained from three pools of words; words of short spoken duration (one syllable words), of medium spoken duration (two syllable words), and of long spoken duration (three and four syllable words). Six words of each length were chosen from children's books used by the youngest group so that they would be familiar to all the subjects. The words used were as follows: egg, pig, bus, car, fish, leaf (short words); rocket, monkey, tiger, tractor, apple, indian (medium words); and fire engine, elephant, umbrella, helicopter, kangaroo, banana (long words). For each word length, 10 lists were constructed by sampling at random without replacement from one of the three sets of words. Each subject therefore received a total of 30 word lists; 10 for each of the three word lengths. The length of the lists had to be suited to the age of the children to avoid "floor" and "ceiling" effects. The 10 and 7 year olds received lists of five words and the 4 year olds lists of four words.

Procedure. Practice trials involving lists of digits were given to all the children to ensure they understood the nature of the serial recall task. In the experiment itself all lists at a given word length were presented successively but the order in which subjects did the three word lengths was determined by a Latin Square. Each list was read to the subject at one word per second and, as soon as the last word had been read, the subject attempted to recall the words in the order presented. They did

this verbally, saying “blank” if they could not recall a particular word, with credit only being given for words occurring in correct serial position. After five lists of one word length, subjects were given a short rest before doing the next five lists. After this serial recall task subjects were given three word pairs and three word triads of a particular word length and asked to repeat each pair and each triad 10 times, continuously as fast as possible. Both word pairs and word triads were used to check that the length of the sequence did not affect the estimates of speech rate obtained. Care was taken, particularly with the younger children, to ensure that they had successfully memorized the triad, which would be repeated to them as necessary before timing repetitions of these triads. Their speech was recorded and a stopwatch was used to measure the duration of repetitions of the words. The times obtained were transformed into speech rate in words per second and averaged for each word length.

Acoustic Similarity

The procedure used was identical in every way to that described above for the word-length study. The materials used here consisted of a set of six acoustically similar words (hat, bat, cat, rat, tap, and mat) and six acoustically dissimilar words (girl, spoon, horse, clock, hand, and train). These words are a subset of the items used by Conrad (1971). The order of word sets was counterbalanced across subjects. Speech rate for each set of words was measured in just the same way as for the word-length study. The order of testing was also balanced for subjects in each group: half doing the acoustic similarity study first, half doing it second. Testing was conducted on 2 consecutive days.

Results and Discussion

The results are shown in Fig. 1 where the mean number recalled per list is plotted as a function of speech rate for each type of word and for each age group. The speech rate scores plotted here are based on the repetition of word triads. The data for the repetition of word pairs were identical in all important respects; the analyses reported here are therefore all based on the word triad data. Each age group contributes five points to this graph. The two filled symbols are for the acoustically similar and acoustically dissimilar items. The three unfilled symbols are for the words of differing lengths: a point for short words (highest speech rate and highest recall for that group), a point for medium length words (lower speech rate and lower recall), and a point for long words (lowest speech rate and lowest recall). The regression line drawn on the graph is the best fitting line for the data from the short, medium, and long words fitted by the method of least squares.

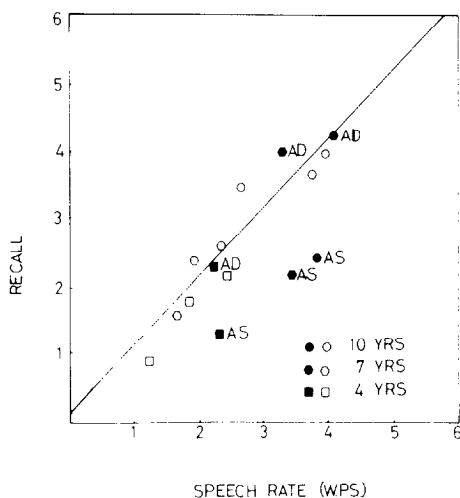


FIG. 1. Mean number of words recalled per list versus speech rate for 10-, 7-, and 4-year-old children. Unfilled symbols denote data for words of different length; filled symbols denote data for acoustically similar (AS) and acoustically dissimilar (AD) words.

Word Length

An analysis of variance was performed on the mean number of words recalled per list in which there were two between-subject factors (age and order of conditions) and one within-subjects factor (word length). This analysis revealed a significant effect of word length ($F(2, 54) = 76.08, p < .001$) and a significant effect of age ($F(2, 27) = 12.12, p < .001$), but no significant interaction between word length and age ($F(4, 54) = 2.23, N.S.$). Tukey HSD tests showed that all pairwise comparisons between the recall of the short, medium, and long words were significant and similarly there were significant differences in recall between all three age groups. No other effects in this analysis were significant.

The same analysis was performed on the speech-rate scores. The results showed a significant effect of word length ($F(2, 54) = 81.01, p < .001$) and a significant effect of age ($F(2, 27) = 8.62, p < .001$), but no significant interaction between word length and age ($F(4, 54) = 2.20, N.S.$). No other effects in this analysis were significant. Tukey HSD tests showed that all pairwise comparisons between the speech-rate scores for the short, medium, and long words were significant and similarly there were significant differences in speech rate between adjacent age groups.

To test the idea that at any age subjects can recall as much as they can say in a fixed time interval it is necessary to show that the slope of the function relating recall to speech rate does not change with age. To this end, the best fitting regression line was fitted to the results of each subject and the slopes were used in a one-way analysis of variance. The

mean slope values were 0.89, 1.18, and 0.90 for the 4, 7, and 10 year olds, respectively; these values do not differ significantly ($F < 1.0$). The intercepts of these regression functions were also checked for any systematic changes with age. The mean intercepts were -0.13 , -0.19 , and 0.84 for the 4, 7, and 10 year olds, respectively; these values also do not differ significantly ($F(2, 33) = 1.76$, N.S.). This allows us to draw a single regression line through the points on Fig. 1. The line was calculated by the method of least squares and has a slope of 0.99 s.

Acoustic Similarity

The recall and speech-rate scores for the acoustically similar and acoustically dissimilar lists plotted in Fig. 1 were analyzed in the same way as the word-length data. An analysis of variance was performed on the mean number of words recalled per list in which there were two between-subject factors (age and order of conditions) and one within-subjects factor (acoustic similarity). This analysis revealed a significant effect of acoustic similarity ($F(1, 30) = 158.15$, $p < .001$), a significant effect of age ($F(2, 30) = 14.57$, $p < .001$), and a significant interaction between acoustic similarity and age ($F(2, 30) = 6.38$, $p < .005$). No other effects in this analysis were significant. The interaction between acoustic similarity and age reflects the fact that the differences in recall between the acoustically similar and acoustically dissimilar words is larger in the 7- and 10-year-old groups than in the 4 year olds; a Tukey HSD test showed that this difference was reliable in all three age groups, however.

The speech-rate data were analyzed in the same way as the recall data. This revealed a significant effect of age ($F(2, 30) = 9.76$, $p < .001$) but no significant effect of acoustic similarity ($F < 1.0$) and no significant interaction between age and acoustic similarity ($F(2, 30) = 1.02$, N.S.). No other effects in this analysis were significant. Thus older children say both types of words more quickly than younger children and there is no reliable difference in the rate of repetition of acoustically similar and acoustically dissimilar words.

Comparison of Word-Length and Acoustic-Similarity Results

In the case of words of different length it is clear in this study, as in previous ones (Hulme et al., 1984), that this variable has consistent effects on recall in children of different ages and, most critically, differences in recall between these words are very closely tied to the rate at which they can be spoken. The results for acoustically similar and acoustically dissimilar items differ in a number of ways from those for word length. It is clear that when expressed as difference scores the size of the acoustic-similarity effect in this study, as in a previous one (Hulme, 1984), increases with age. A new finding here, however, is the relationship

between acoustic similarity and speech rate. Differences in recall between acoustically similar and acoustically dissimilar words cannot be accounted for by differences in speech rate. The two categories of words are repeated at essentially identical rates but the acoustically similar items are recalled much more poorly. So this pattern of results clearly contradicts the hypothesis outlined in the introduction that differences in recall between acoustically similar and dissimilar words may reflect differences in speech rate for these two classes of words.

While the scores for the acoustically dissimilar words are well fit by the same regression function relating recall to speech rate for the words of different lengths, recall for the acoustically similar words is depressed below this function. This was confirmed in the following way. For each subject the results for recall and speech rate for words of different lengths were fitted by a regression analysis. Now given that for each subject a speech rate for the acoustically similar and acoustically dissimilar words was available, it is possible to derive an expected recall value based on those speech rate values, using the regression function for different word lengths. Having done this, the observed and expected recall scores for the acoustically similar and acoustically dissimilar words were compared in an analysis of variance where the factors were age, type of word, and observed/expected scores. The crucial result here is that there was a highly significant interaction between the factors of observed/expected and type of word (acoustically similar or acoustically dissimilar). This interaction reflects the fact that the observed and expected recall scores for the acoustically dissimilar words were very similar, whereas for the acoustically similar items the observed recall scores were much lower than those predicted on the basis of their speech rate. Tests for simple main effects using a pooled residual error term (Kirk, 1982) showed there was a highly significant discrepancy between the observed and expected recall values for the acoustically similar words ($F(1, 99) = 33.04, p < .001$) but there was no significant discrepancy between the observed and expected recall scores for the acoustically dissimilar words ($F(1, 99) < 1.0$).

One other effect apparent in Fig. 1 is that the two older groups who speak faster show a larger absolute difference between the recall of acoustically similar and acoustically dissimilar words than do the 4 year olds. To explore this association between speech rate and the effects of acoustic similarity more thoroughly some correlations were computed. As a measure of the size of the acoustic similarity effect the absolute difference in recall between the two types of words was computed for each subject. The correlation between the acoustic similarity effect and speech rate is substantial ($r = .54, p < .01$), but speech rate itself correlated highly with age ($r = .61, p = .01$) and the correlation between the size of the acoustic similarity effect and age was ($r = .39, p < .05$).

However, the partial correlation between the acoustic similarity effect and speech rate, partialling out the effects of age, remains quite substantial ($r = .41, p < .02$). In contrast, the partial correlation between the acoustic similarity effect and age partialling out the effects of speech rate was very near to zero ($r = .09, N.S.$). Thus, it can be argued quite strongly that the important link is between the acoustic similarity effect and speech rate rather than age. This is an important result, since, if we interpret speech rate as a measure of rehearsal speed, it ties the size of the acoustic similarity effect to this rehearsal process.

EXPERIMENT 2

One possible criticism of the findings relating the effects of acoustic similarity to speech rate in Experiment 1 is the influence of range effects. In particular the recall of acoustically similar words by the 4 year olds is at a very low level. It might be, therefore, that the magnitude of the acoustic similarity effect is artificially reduced in this group. Because of the theoretical importance of this result it was decided to conduct a replication. In order to avoid the possible influence of floor and ceiling effects in any of the age groups a memory span procedure was adopted instead of the fixed list length procedure used in Experiment 1.

Method

Subjects. A total of 36 children participated in the study; twelve 5 year olds (mean age 5 years 3 months; range 5 years 0 months to 5 years 8 months), twelve 7 year olds (mean age 7 years 9 months; range 7 years 4 months to 7 years 10 months), and twelve 9 and 10 year olds (mean age 9 years 10 months; range 9 years 2 months to 10 years 4 months). There were equal numbers of boys and girls in the two older groups and nine boys and three girls in the 4-year-old group. The children came from two different schools both of which had predominantly middle-class intakes.

Materials. In order to relate recall to speech rate, measures of both variables were obtained from two pools of words. One pool consisted of eight acoustically similar words (hat, mat, rat, cat, bat, bag, tap, man) and the other of eight acoustically dissimilar words (clock, fish, horse, spoon, train, hand, girl, bus). All the words are items used by Conrad (1971). For each of the two word pools, eight lists were constructed by sampling at random without replacement at successive list lengths ranging from two to seven words.

Procedure. Practice trials involving lists of digits were given to all the children, to ensure they understood the nature of the serial recall task. Two determinations of memory span were made for each set of words. Four lists at each list length, starting with the two word lists, were presented in ascending order. Each list was read to the subject at a rate

TABLE 1
MEAN MEMORY SPAN SCORES (AND STANDARD DEVIATIONS) FOR ACOUSTICALLY SIMILAR AND
ACOUSTICALLY DISSIMILAR WORDS AS A FUNCTION OF AGE IN EXPERIMENT 2

Word type	Age		
	5	7	10
Acoustically similar	3.19 (0.53)	3.21 (0.27)	3.40 (0.48)
Acoustically dissimilar	3.89 (0.82)	4.00 (0.48)	4.54 (0.64)

of one word per second and as soon as the last word had been read, the subject attempted to recall the words in the order presented. Presentation of further lists continued until the subject failed to recall, in the correct serial order, three out of the four lists at a given length. At this point, the word lists of the other type (similar or dissimilar) were presented until the subject again failed to recall three out of four lists at a given list length correctly. At this point, another set of lists was presented, starting at the list length for each word type where the subject had been completely correct previously. Once again, list length was increased until three out of the four lists at a given length were incorrectly recalled. The order of conditions was counterbalanced across subjects in each group.

After the serial recall task, speech rate was measured by giving subjects four randomly constructed word pairs for each category, to repeat 10 times each, continuously as fast as possible. A stopwatch was used to measure the duration of these repetitions. The mean of the four times for each category was calculated and transformed into speech rate in words per second.

Results and Discussion

Recall

A memory span score was computed as the longest list length for which the subject recalled all four lists completely correctly, plus 0.25 for each single longer list that was correctly recalled. These scores, averaged across both determinations, are shown in Table 1.

An analysis of variance was performed on these scores in which there were two between-subject factors (age and order of conditions) and one within-subjects factor (acoustic similarity). This revealed a significant effect of acoustic similarity ($F(1, 30) = 228.26, p < .001$), a marginally significant effect of age ($F(2, 30) = 3.13, p < .058$), and a highly significant interaction between age and acoustic similarity ($F(2, 30) = 8.16, p < .01$). No other effects in this analysis were significant.

TABLE 2

MEAN SPEECH RATE FIGURES IN WORDS PER SECOND (AND STANDARD DEVIATIONS) FOR ACOUSTICALLY SIMILAR AND DISSIMILAR WORDS AS A FUNCTION OF AGE IN EXPERIMENT 2

Word type	Age		
	5	7	10
Acoustically similar	1.99 (0.51)	2.21 (0.32)	2.72 (0.38)
Acoustically dissimilar	2.13 (0.53)	2.36 (0.28)	2.99 (0.36)

The interaction between age and acoustic similarity clearly shows that as age increases acoustic similarity has progressively more effect on recall. To check on this interpretation difference scores between spans for acoustically similar and dissimilar words were computed. There was a significant linear trend in these scores across the three age groups ($F(1, 30) = 22.00, p < .01$).

Speech Rate

The average speech rates, in words per second, are shown in Table 2.

An analysis of variance on these scores revealed a significant effect of age ($F(2, 30) = 13.01, p < 0.001$) and a significant effect of acoustic similarity ($F(1, 30) = 20.25, p < 0.001$), but no significant interaction ($F(2, 30) = 1.42, N.S.$). No other effects in this analysis were significant. Thus older children repeat both types of words more quickly than younger children, and all age groups repeat the acoustically dissimilar words slightly more quickly than the acoustically similar words.

To investigate the relationship between the increase in the size of the acoustic similarity effect and changes in speech rate with age some correlations were computed. The correlation between the size of the acoustic similarity effect and speech rate was significant ($r = 0.53, p < .01$). Speech rate itself was again highly correlated with age ($r = 0.64, p < .01$) and the correlation between age and the size of the acoustic similarity effect was ($r = 0.45, p < .01$). Most crucially, the partial correlation between the size of acoustic similarity effect and speech rate, partialling out the effects of age, was again significant ($r = 0.36, p < .05$). In contrast, the correlation between the size of the acoustic similarity effect and age, partialling out speech rate, was again found to be not significant ($r = 0.17, N.S.$). This pattern of results provides confirmation of the relationship between the size of the acoustic similarity effect and speech rate found in Experiment 1. This is an important finding and apparently a robust one. It should be noted that difference scores for

correlated measurements such as the acoustically similar and dissimilar span scores, are inherently less reliable than the scores from which they are derived (because error variances of the two raw scores are additive). The correlations obtained here involving the effects of acoustic similarity are, therefore, probably conservative.

One minor difference in the results of this experiment compared to those of Experiment 1, is that a significant difference in speech rate for the acoustically similar and acoustically dissimilar words was found. This difference is small in magnitude (6%) and cannot account for the large difference in recall between these two sets of words. The discrepancy between the two experiments here presumably reflects the difference in word sets used: two pools of six words were used in Experiment 1 while two additional words were included in each pool for the present experiment.

GENERAL DISCUSSION

These experiments have generated a number of findings. Word length has consistent effects on serial recall across a wide range of ages, and the recall of words of different lengths is closely related to the rate at which they can be spoken. This replicates our earlier findings (Hulme et al., 1984). Acoustic similarity has more effect on recall as age increases and changes in the size of this effect are also closely related to changes in speech rate with age.

The explanation which is favored for these results is based closely on the idea of Baddeley and Hitch's (1974) articulatory loop. According to the original formulation of their working memory model, the articulatory loop could store a certain amount of speech in an articulatory code. Information in this store was subject to passive decay over time, and, in order to overcome this, rehearsal involving the reactivation of articulatory or speech motor programs took place. Thus this loop was in some ways analogous to a recirculating tape loop, in which items were constantly being read and rewritten to overcome the effects of decay.

A major piece of evidence which influenced the formulation of this view was the effect of word length on serial recall. The articulatory loop provides a simple explanation for this effect. Longer words are stored in terms of articulatory commands of longer duration than short words (as indexed by how quickly they can be spoken). This has the effect of limiting the number of words which can be rehearsed before information has decayed too far for it to be recognized: fewer long words than short words can be rehearsed within the decay time of the store.

This model accounts very well for the developmental data concerning word length that have been presented here. As children get older their speech rate increases. This in turn increases the speed with which items in the loop can be rehearsed to an equal extent for words of different

lengths, and this leads to a proportionally equal increase in recall for words of different lengths.

The data for acoustic similarity can be accommodated within this model in the following way. Acoustically similar items will inevitably lead to the setting up of highly similar traces in the articulatory loop. It therefore follows that the traces for such sequences of items will be more difficult to discriminate one from another, and, in a system which is subject to loss of information due to decay over time, any loss of information will affect acoustically similar items more seriously than acoustically dissimilar items.

A reasonable assumption would be that errors are introduced at the point at which items in the store are "read" before the appropriate articulatory code is reactivated. Errors are introduced at this stage by guessing on the basis of partial information. Now, as age and so speech rate increases, more items will be held in the loop, and inevitably more items will be read and rewritten into the store per unit of time. Because there are different base error rates for the acoustically similar compared to the acoustically dissimilar items, as speech rate increases, the absolute difference in recall between acoustically similar and acoustically dissimilar lists will increase. So although speech rate for the acoustically similar words goes up substantially with age the increase in recall of these words does not show an equivalent increase because the probability of errors in identifying memory traces of these items increases.

This view clearly implies that although the difference in recall between acoustically similar and dissimilar words will increase in absolute terms, in proportional terms the difference should remain roughly constant. To check on this the proportion of words correct for the similar and dissimilar lists in Experiment 1 were subjected to an analysis of variance. This revealed a significant effect of acoustic similarity ($F(1, 30) = 118.95$; $p < .001$) and a significant effect of age ($F(2, 30) = 5.05$; $p < .05$), but most crucially no interaction between these two factors ($F(2, 30) = 2.19$; N.S.). This supports the idea that the confusion between items in the acoustically similar lists is operating with the same force in different age groups. However, because older subjects rehearse more items within a given unit of time their absolute losses on the acoustically similar lists relative to the acoustically dissimilar lists is greater than in younger subjects.

The present results provide support for the importance of changes in speech rate with age as an explanation of developmental improvements in short-term memory performance. It is important to note that for the words of different lengths and for the acoustically dissimilar words there is a very close quantitative relationship between how quickly the words can be spoken and how well they are recalled. Thus all age groups recall a roughly constant amount of these words when expressed in terms of

how long it takes to say them. This impressively close link between speech rate and recall explains both differences between subjects (age effects) and differences between materials. These findings are readily explained by reference to the concept of an articulatory loop which holds information in the form of speech motor programs.

This view of short-term memory development is one example of an explanation that stresses the importance of changes in the efficiency of basic or nonstrategic information-processing operations or representational capacities (cf. Dempster, 1981, 1985). The most prominent alternative explanation of this sort was provided by Case, Kurland, and Goldberg (1982). They measured the time taken by children of different ages to identify spoken words by measuring how long it took them to begin to repeat a word played to them on a tape recording. They found that this time decreased between the ages of 4 and 6 and that speed on this measure correlated with memory-span scores. This result was interpreted in terms of a trade-off between attentional capacity allocated to operations such as item identification and capacity allocated to storage, the less efficient operations in younger children leaving less capacity available for storage.

We have argued before (Hulme et al., 1984) that the repetition of individual words or small groups of words as measured in the present experiments provides a measure of speech rate which is independent of the time taken for item recognition. In contrast the measure used by Case et al. (1982) could be viewed as a composite of item-recognition time and speech-onset time. The fact that we find such a substantial relationship between memory span and speech rate seems to show that the item-identification component of the measure used by Case et al. may not be the relevant determinant of changes in span with age. A similar argument to this has also recently been advanced by Huttenlocher (1984). Instead our results point quite clearly to changes in the efficiency of articulation with age as being the important factor in accounting for developmental increases in memory span.

An alternative argument however might be to suggest that the effects of word length observed here, and in other experiments, actually reflect differences in item identification speed. If long words take longer to identify this may explain why they are less well recalled (Dempster, 1981). Such a view does not explain why there should be such a close link between speech rate and recall but perhaps this could be accommodated if it was assumed that differences in item-identification speed closely parallel differences in speech rate. The most direct test of these alternatives was presented by Hitch, Halliday, and Littler (1983). They measured memory span for one, two, and three syllable words in 8- and 11-year-old children. They also obtained measures of articulation rate and word-identification time. Their results showed that articulation rate was an excellent predictor of memory span for the different materials as it was

in the present study. Item identification time proved to be a much poorer predictor of memory span. Hitch et al. (1983) also showed that articulatory suppression, which is known to block rehearsal in the articulatory loop, abolished the effects of word length on memory span. We have also obtained such a result with presentation of pictures to children between the ages of 4 and 10 (Hulme, Silvester, Smith, & Muir, 1986). These results fit well with the idea of an articulatory rehearsal process underlying the effects of word length but would be difficult to reconcile with the idea that word-length effects are mediated by differences in the speed of item identification.

There is an obvious conflict between the present explanation for the effects of acoustic similarity and the most recent refinement of the working memory model proposed by Baddeley (1986; Salame & Baddeley, 1982). According to this view the articulatory loop is now divided into two components, a phonological input store and an articulatory rehearsal process. In this latest version of the model auditory speech input gains obligatory access to the phonological store, while visual information is only input via subvocal rehearsal. The effects of acoustic similarity are attributed to the laying down of similar and so confusable traces in the phonological store. The effects of word length are again attributable to the articulatory rehearsal process.

The main difference between this view and the one advanced above, based upon the earlier conception of the articulatory loop, centers on the necessity of postulating a phonological input store. The main reason for advancing this concept is the unattended-speech effect (Salame & Baddeley, 1982); memory for visually presented words is impaired by the simultaneous presentation of spoken material which the subject is told to ignore. In the framework outlined above, this unattended-speech effect can be explained simply by assuming that auditory speech input gains obligatory access to the articulatory loop. The simultaneous input of items in the unattended speech experiments simply leads to problems of encoding the visually presented material onto the loop (see Broadbent, 1984). The fact that this effect is abolished by articulatory suppression can be explained in terms of suppression blocking access of visually presented material to the articulatory loop. This also explains why with visual presentation articulatory suppression abolishes the effects of word length (Baddeley et al., 1975) and acoustic similarity (see, e.g., Crowder, 1978).

One feature of the current position is that the occurrence of acoustic-similarity effects is very directly tied to the process of rehearsal. This is consistent with other evidence that the effects of acoustic similarity are most in evidence in situations where the opportunities for rehearsal are greatest (Crowder, 1978); for example, Conrad (1967) showed that the size of the acoustic-similarity effect was decreased and eventually

abolished if subjects were required to shadow irrelevant digits between presentation and recall. Recently, further evidence converging on the importance of rehearsal in accounting for the acoustic-similarity effect and changes in this effect with age have been presented by Cowan, Cartwright, Winterowd, and Sherk (1987). These authors also set out to investigate the change in the size of the acoustic similarity effect with age reported by Hulme (1984). They found, in a number of experiments, that adults who recalled lists of spoken words following articulatory suppression that blocked rehearsal, displayed patterns of recall similar to those of 5-year-old children. This of course is entirely consistent with the present position. Furthermore, Cowan et al. (1987) found that blocking rehearsal was more detrimental to recall of acoustically dissimilar than similar lists. This, in a sense, is the converse of what we found in the present study: that increases in the speed of rehearsal across age groups were associated with larger increases in recall for the acoustically dissimilar than the similar lists.

As a final point, it is hoped that the present results are one demonstration of the potential value of developmental data, as a means of testing and refining models of cognitive processes in adults (see Hulme, 1986). Whether or not the present explanation for the differing developmental patterns in the effects of acoustic similarity and word length are accepted, it is clear that any adequate model designed to account for these effects must be able to account for these developmental differences also.

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